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Beyond the hype: unpacking Italian university students' engagement with generative AI for learning and academic integrity

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The rise of Generative Artificial Intelligence (GAI) is reshaping higher education by influencing how university students access, process, and apply knowledge. This study explores university students' perceptions and usage patterns of GAI, with a specific focus on ChatGPT, through a large-scale survey of 1,366 students from 24 Italian universities. Employing cluster analysis, the research identifies four distinct student groups based on their digital skills and attitudes toward GAI: Indifferent, Fearful, Pragmatics, and Enthusiasts. Findings indicate significant variations in students' engagement with GAI, with factors such as gender, degree programs, and technological proficiency shaping their perspectives. While some students embrace AI as a valuable educational tool, others express ethical and academic integrity concerns. Notably, digital competence correlates with AI adoption, with STEM students exhibiting higher engagement. The study highlights the need for universities to address both the opportunities and challenges of GAI, proposing evidence-based policies to foster responsible integration into academic settings. By providing a nuanced understanding of student interactions with AI, this research aims to inform pedagogical strategies and regulatory frameworks to balance innovation with ethical considerations in higher education.

Keywords: GAI; artificial intelligence; higher education; university; students

Introduction

The advent of Generative Artificial Intelligence (GAI), a specialized branch of Artificial Intelligence (AI), represents a turning point in the evolution of contemporary higher education, fundamentally reshaping the manner in which knowledge is accessed, processed, and applied in academic contexts (Rahiman and Kodikal 2024). While the formal origins of AI research are often traced back to the Dartmouth workshop of 1956, led by computer scientist John McCarthy (Skinner 2012), AI's practical applications remained largely unnoticed by the general public for much of its history. However, the launch of ChatGPT (Chat Generative Pre-trained Transformer) in November 2022 marked a watershed moment. Within just a week, ChatGPT attracted over a million users (Dempere et al. 2023), catalysing extensive discourse on the societal, ethical, and economic implications of GAI (Hagendorff 2024).

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The integration of GAI into higher education presents a complex landscape of opportunities and challenges. On one hand, these tools offer new possibilities for personalized learning, accessibility, and efficiency in knowledge acquisition. On the other hand, they raise fundamental questions about student engagement with learning processes, the development of critical thinking skills, and the boundaries of academic integrity. The capacity of GAI to generate human-like content has blurred traditional distinctions between student-produced work and AI-assisted or AI-generated submissions, creating ambiguity in academic settings (Yusuf, Pervin, and Román-González 2024). Moreover, perceptions of appropriate GAI use vary significantly across cultural contexts and academic disciplines, reflecting diverse perspectives on what constitutes legitimate educational support versus academic misconduct (Yusuf, Pervin, and Román-González 2024).

Public sentiment towards AI tools reveals a diverse spectrum of perspectives ranging from optimism to apprehension, as evidenced by a recent global survey conducted by IPSOS (2024). The survey reveals this dichotomy, with approximately half of respondents expressing enthusiasm for such technologies, while an equivalent proportion reporting feelings of anxiety. A particularly noteworthy finding is that 46% of Generation Z participants anticipate that AI will replace their current employment within the next five years. Moreover, 66% of respondents expect a significant impact on their daily lives in the near future, while 60% foresee substantial changes in their professional activities within the same timeframe.

This study investigates these perceptions more deeply among university students. Understanding student perspectives on GAI is crucial, given that their attitudes may vary based on academic discipline, technological proficiency, and ethical considerations surrounding AI use. Their perspectives and usage patterns have profound implications for teaching and learning outcomes, academic integrity practices, and the broader societal role these technologies may play in the future (Chan and Zhou 2023). By elucidating these diverse interactions, this research aspires to inform the development of policies and practices that not only harness the benefits of GAI but also effectively address its challenges. Although empirical studies on the use and perception of AI tools in education are increasingly available (Bond et al. 2024; Wang, Jing, and Shen 2025), there remains a paucity of research on GAI within the Italian higher education system. To address this gap, we conducted a large-scale quantitative study involving 1,366 participants from 24 academic institutions across Italy. By employing cluster analysis, this research aims to categorize students surveyed into distinct groups based on their usage patterns and perceptions, thereby illuminating the nuances of their experiences with GAI. The research was guided by the following questions:

1. RQ1: What patterns of adoption do Italian university students exhibit regarding GAI tools, and how do these patterns differ across socio-demographic factors, personal characteristics, degree programs, and levels of digital skills?
2. RQ2: How do Italian university students perceive the role of GAI in their academic and personal activities?
3. RQ3: What are the varying perspectives among students on the long-term implications of GAI for their career trajectories, academic growth, and broader societal transformations?
4. RQ4: What ethical and regulatory concerns do Italian university students express regarding the use of GAI?

This manuscript is organized into four sections. The first section offers a comprehensive literature review synthesizing theoretical frameworks and empirical evidence related to student perspectives on GAI as an educational tool. The second section delineates the methodological approach, detailing the analytical procedures used and describing the demographic and academic characteristics of the study sample. The third section presents an in-depth analysis of survey results across four clusters, integrating both quantitative data and qualitative findings for each group. Finally, the fourth section discusses broader implications for educational policy formulation and pedagogical innovation, while identifying avenues for future research.

Literature review: exploring students' views on GAI in higher education

While the emergence of GAI is a recent phenomenon, the use of artificial intelligence in educational settings has long attracted scholarly attention. This interest has historically focused on adaptive teaching strategies, personalized learning approaches, and the integration of AI technologies in classrooms (for an exhaustive literature review see Chiu et al. 2023). However, the current landscape has shifted significantly with the widespread adoption of ChatGPT and similar GAI tools, leading to evolving meanings and usage practices that prompt diverse responses ranging from admiration to scepticism (Montenegro-Rueda, Fernández-Cerero, and Fernández-Batanero 2023). Researchers and educators are now confronted with critical questions regarding the effective and ethical implementation of AI, its long-term implications for teaching roles, and the safeguarding of educational integrity and authenticity (Kramm and McKenna 2023; Popenici 2023).

Students' engagement with generative AI in higher education

The integration of GAI into students' learning practices reveals a complex tapestry of engagement patterns that simultaneously enhance and potentially undermine traditional pedagogical objectives. Existing literature suggests that students possess a nuanced understanding of GAI's potential benefits and challenges in educational applications (Meade, Ogundamisi, and Tiratelli 2023), though the depth and nature of this engagement vary considerably across contexts. Many studies indicate a generally positive attitude among students towards utilizing applications like ChatGPT as valuable learning tools (e.g. Karakose and Tülübaş 2023). These tools are perceived as beneficial for enhancing student engagement, motivation, and personalized learning experiences that cater to diverse needs and learning styles (Cotton, Cotton, and Shipway 2024). The personalization dimension is particularly significant: GAI tools foster self-directed learning by enabling students to explore topics at their own pace while receiving immediate clarification on complex subjects (Holmes and Tuomi 2022). Furthermore, these technologies provide continuous feedback on performance along with tailored recommendations (Chan and Hu 2023), strengthening students' connection to the subject matter and improving knowledge retention. Beyond personalization, students leverage GAI across multiple functional domains. Non-native speakers have utilized GAI as writing assistants to enhance their text-editing skills by addressing grammatical and structural issues (Erito 2023; Hasanein and Sobaih 2023). Applications like ChatGPT facilitate proofreading, summarizing arguments, condensing content, and paraphrasing, though perceptions regarding ethical acceptability vary widely (Črček and Patekar 2023; Farhi et al. 2023). Students also utilize GAI technologies for brainstorming ideas and generating content that alleviates writing pressure and reduces cognitive load (Al-Obaydi, Pikhart, and Klimova 2023).

Notably, the literature suggests that AI has a transformative effect on student analytical skills, enhancing their ability to evaluate information and formulate critical inquiries (Kosmyna et al. 2025). Most students acknowledge AI's significant influence on their academic fields and future careers while expressing a desire for AI integration in their educational pursuits (Lee, Hwang, and Chen 2022). They consider this integration an indispensable component of university curricula (Abdelwahab, Rauf, and Chen 2022).

However, these affordances are counterbalanced by significant concerns about cognitive development and learning depth. A prominent concern pertains to the potential impact of AI on students' intellectual growth, raising questions about whether such technologies serve to augment or impede their development (Zhang 2024). Some students worry about excessive reliance on technology for academic tasks, which may potentially result in a decline in confidence in problem-solving abilities, as well as hinder the development of critical thinking skills and creativity (Kayalı et al. 2023; Ngo 2023). In some cases, students report discomfort or anxiety when access is restricted (Ajlouni, Abd-Alkareem Wahba, and Salem Almahaireh 2023), suggesting a level of dependency that may compromise learning autonomy.

The tension between efficiency and authentic learning engagement represents a critical fault line in current debates. While GAI can reduce cognitive load for routine tasks, thereby freeing mental resources for higher-order thinking, it may simultaneously create what Kosmyna et al. (2025) term 'cognitive debt', a superficial engagement with material that undermines deep learning processes. This paradox highlights the need to distinguish between GAI use that scaffolds learning versus use that substitutes for intellectual effort. A prevalent concern among students is also the potential reduction of human interaction in the learning process along with a lack of emotional intelligence and empathy in AI-mediated exchanges (Bisdas et al. 2021). Idroes et al. (2023) found that over one-third of undergraduate respondents expressed apprehension about AI's impact on interpersonal relationships within educational settings, fearing that AI integration could diminish educational quality by reducing meaningful interactions between students and instructors. Similarly, Seo et al. (2021) highlighted Canadian student concerns regarding AI potentially weakening human-to-human interactions, which are crucial for fostering deeper engagement and emotional support. Concerns have also been raised about potential reductions in job opportunities resulting from widespread AI adoption across various professional fields, alongside increased demand for specific job skills due to technological advancements, a sentiment reflected in broader societal anxieties regarding AI's role in future work environments (Ghotbi, Ho, and Mantello 2022).

Disciplinary and contextual differences in GAI adoption

The adoption and perception of GAI tools are not uniform across academic disciplines or cultural contexts, reflecting fundamental differences in epistemological traditions, pedagogical practices, and technological readiness. Previous surveys have revealed variations in university students' attitudes towards technology adoption based on their field of study. Engineering and architecture students tend to exhibit a greater inclination towards new technologies compared to their peers in social sciences and humanities (Huedo-Martínez, Molina-Carmona, and Llorens-Largo 2023). This disparity underscores the significant influence of familiarity with technology and its perceived relevance to one's discipline on adoption behaviors. STEM disciplines, with their emphasis on computational thinking and problem-solving, may find more immediate and instrumental applications for GAI compared to humanities fields where critical interpretation, rhetorical nuance, and original

argumentation are paramount. However, this does not necessarily imply that GAI is less valuable in humanities contexts; rather, it suggests that patterns of integration and ethical considerations may differ substantially across disciplinary boundaries.

Global shifts in how these tools are utilized reflect cultural beliefs, social values, and regulatory differences (Acosta-Enriquez et al. 2024; Roberts et al. 2023), making it essential to examine national contexts. Cultural dimensions, such as attitudes toward authority, individualism versus collectivism, and tolerance for uncertainty, may shape how students perceive the legitimacy of AI assistance and the boundaries between acceptable support and academic misconduct. Moreover, national regulatory frameworks and institutional policies create different opportunity structures for GAI adoption. For instance, countries with more developed AI governance frameworks may foster more transparent discussions about appropriate use, while contexts with regulatory voids may leave students navigating ethical ambiguities without institutional guidance. Although research on GAI tools such as ChatGPT has grown rapidly, there remains a lack of in-depth studies that capture students' lived experiences and nuanced perspectives in specific national contexts, particularly within Italy. The Italian higher education system, with its distinctive characteristics, including strong disciplinary traditions, varying degrees of digital infrastructure across institutions, and specific cultural norms around academic work, represents an important yet under-researched context. Understanding Italian students' engagement with GAI not only fills a geographical gap in the literature but also provides comparative insights that can illuminate how broader European educational values and practices intersect with emerging AI technologies.

Ethical concerns and academic integrity in GAI-mediated learning

The capacity of GAI to generate human-like content has blurred traditional distinctions between student-produced work and AI-assisted or AI-generated submissions, creating ambiguity in academic settings and raising fundamental questions about authorship, originality, and intellectual honesty (Yusuf, Pervin, and Román-González 2024). A significant challenge arises from the potential for these tools to produce content indistinguishable from human-generated work (Sullivan, Kelly, and McLaughlan 2023), leading to serious ethical concerns surrounding dishonesty and misuse, such as cheating or plagiarism, especially when students employ these tools without proper attribution (Rawas 2024). The traditional concept of plagiarism, predicated on the unauthorized use of another person's work, becomes conceptually murky when the 'author' is an algorithm trained on vast corpora of human-generated text. Moreover, perceptions of appropriate GAI use vary significantly across cultural contexts and academic disciplines, reflecting diverse perspectives on what constitutes legitimate educational support versus academic misconduct (Yusuf, Pervin, and Román-González 2024). This ambiguity is compounded by the fact that many students advocate for clearer guidelines from academic institutions alongside comprehensive training on appropriate tool usage within educational activities (Tierney, Peasey, and Gould 2025). However, Malmström, Stöhr, and Ou (2023) found that most students were unaware if their institutions had already established regulations governing ethical AI tool use, suggesting a troubling disconnect between institutional policy development and student awareness.

Concerns have also been raised about the reliability of the algorithms underlying these tools. GAI can generate inaccurate information or misleading content; for example, ChatGPT has been found to produce incorrect citations or fictitious references that undermine research validity (Farrokhnia et al. 2024; Rudolph, Samson, and Shannon 2023). The

inability to evaluate source credibility combined with reliance on AI-generated content without proper verification risks propagating misinformation, an especially troubling issue in educational contexts where knowledge pursuit is paramount (Famaye et al. 2024). Students have also expressed concerns about potential biases in GAI models trained on extensive datasets containing imperfections or prejudiced associations. Such biases may lead to distorted responses that reinforce stereotypes about certain demographic groups (Yu 2024), raising questions not only about the accuracy of AI outputs but also about their potential to perpetuate social inequalities. These ethical concerns create a fundamental tension between innovation and integrity. On one hand, restricting GAI access may impede students' preparation for AI-saturated professional environments and limit opportunities for developing critical AI literacy. On the other hand, unfettered access without clear ethical frameworks risks normalizing practices that undermine the core educational values of authentic intellectual engagement and original thought. Navigating this tension requires moving beyond binary choices of permission or prohibition toward more nuanced approaches that integrate ethical reasoning into GAI pedagogy.

Theoretical perspectives on student adoption of GAI

Understanding student acceptance and utilization of GAI requires theoretical frameworks that can explain the cognitive, affective, and social dimensions of technology adoption. The Technology Acceptance Model (TAM), initially proposed by Davis (1989), provides a valuable framework for understanding student acceptance of AI chatbots in educational contexts. According to TAM, attitudes towards new technologies are primarily influenced by two key factors: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). PU refers to the extent to which individuals believe a technology can enhance their performance (Yilmaz, Maxutov, and Baitekov 2023; Saif et al. 2024), while PEOU pertains to the perceived effort required to utilize the system (Lai, Cheung, and Chan 2023; Ma et al. 2024). This model suggests that an individual's intention to adopt new technologies is largely driven by their perception of usefulness; if a technology is regarded as beneficial, individuals are more likely to engage with it. Within an educational context, student perceptions, concerns, and experiences with technological innovations significantly influence their willingness to adopt these tools and integrate them into their learning processes. Research indicates that the diffusion of various technological applications in education is largely driven by factors such as low complexity, perceived advantages, compatibility with existing practices and beliefs, and the ability to address learner and educator needs (Pinho, Franco, and Mendes 2021).

However, while TAM provides valuable insights, its adequacy for capturing the full complexity of GAI adoption warrants critical examination. TAM's emphasis on perceived usefulness and ease of use may underestimate the role of ethical considerations, emotional responses, and social norms in shaping student engagement with GAI. Students' decisions are not purely utilitarian calculations but are influenced by moral intuitions about academic honesty, concerns about cognitive development, and perceptions of legitimacy within their disciplinary communities. Moreover, TAM's individualistic focus may obscure the collective and contextual dimensions of technology adoption, as students' GAI use occurs within institutional environments characterized by specific policies, peer norms, and instructor expectations. Alternative frameworks offer important extensions: the Unified Theory of Acceptance and Use of Technology (UTAUT) incorporates social influence and facilitating conditions alongside performance expectancy and effort expectancy, providing a more comprehensive view of technology adoption in

organizational contexts (Venkatesh et al. 2003; Menon and Shilpa 2023). Similarly, the Expectancy-Value Theory (EVT) employed by Chan and Zhou (2023) recognizes that students weigh not only utility but also the value they place on learning outcomes, the effort required, and the emotional costs or benefits. Yet even these extended models struggle to fully capture the ethical dimensions distinctive to GAI in education, particularly the tension between efficiency and academic integrity, and the distinction between adoption and appropriate use. What emerges is the need for more integrative frameworks that combine cognitive-rational perspectives with ethical reasoning, emotional dimensions, and contextual factors, acknowledging that students are not merely technology users but learners navigating moral ambiguities and developmental challenges.

The preceding review reveals critical gaps in current research. First, while GAI adoption is expanding globally, national contexts like Italy remain under-examined, limiting our understanding of how cultural and institutional factors shape student perceptions. Second, existing studies tend to examine engagement patterns, ethical concerns, or theoretical adoption models in isolation, whereas the intersection of these dimensions remains largely unexplored. Third, research must move beyond simple dichotomies of acceptance versus rejection, as students' attitudes are more nuanced and multidimensional. The present study addresses these gaps both conceptually and empirically: conceptually, by integrating engagement, ethics, and context through a multidimensional framework; empirically, by providing large-scale evidence from Italian universities using cluster analysis to identify distinct student profiles based on digital skills, attitudes toward GAI, and ethical reasoning.

Method and study participants

To investigate the engagement, perceptions, and usage patterns of GAI tools, such as ChatGPT, among Italian university students at various stages of their academic journey, a quantitative research design was employed. An online survey was designed following best practices for online survey development (Sue and Ritter 2007) to efficiently reach a large cohort of students enrolled in undergraduate, master's, and doctoral programs across Italian higher education institutions. The survey comprised both closed-ended and open-ended questions, structured into five sections: (1) demographic and university affiliation; (2) self-reported digital competence and usage patterns of various devices; (3) engagement with GAI tools for personal activities and academic tasks; (4) perspectives on the integration of GAI tools within higher education; and (5) future expectations regarding GAI's impact on society, the labor market, and personal life trajectories.

Before formal data collection, a pre-test was conducted with a small student sample in a controlled laboratory environment to refine the questionnaire's clarity and accuracy. This feedback led to modifications enhancing the questionnaire's reliability and validity. The refined instrument was subsequently administered in a pilot study at the University of Udine to further improve the tool and gather initial interpretations of the investigated phenomenon (Farinosi and Melchior 2025). Based on these two previous phases, the definitive questionnaire was implemented in the present national-scale survey.

The final tool comprised 55 variables, which were organized into closed-ended questions (six of which were evaluated on a Likert scale, two were binary and four were multiple choice) and two open-ended questions. The latter were designed to gather further insights and perspectives on the advantages and disadvantages associated with the spread of generative AI in universities and society as a whole.

The sample

Following Institutional Review Board (IRB) approval, an invitation to participate in the study, accompanied by a research prospectus, was disseminated during the first semester of the 2023–2024 academic year, specifically one year after the public release of ChatGPT. The invitation was extended to the Public Relations Offices (URPs) of all 97 Italian universities, including 11 online institutions (MUR 2024), with a request to distribute the survey link, contingent upon approval by their respective university authorities, to all students enrolled in bachelor's, master's and doctoral programs through their official mailing lists. This strategy resulted in a convenience sample (Galloway 2005) based on students' availability and voluntary participation.

Data were collected from 1,366 university students across diverse disciplines and academic levels between 3 October and 18 December 2023. The majority of them came from 24 Italian universities, with a concentration from six main institutions: Polytechnic of Turin, University of Parma, E-campus, University of Udine, University of Salerno, and IUSVE. Table 1 presents the demographic and academic characteristics of the sample.

Index construction

As a preliminary step, two composite indices were constructed to capture key dimensions of students' relationship with digital technologies and AI. The digital competence index was obtained by summing the values from questions concerning self-declared digital competence and frequency of use of smartphones, personal computers, and search engines. This resulted in a continuous index ranging from 8 to 16 ($M = 13.96$; $Me = 14.00$; $Mo = 15.00$; $SD = 1.45$). The low standard deviation, clustered around a high mean, indicates a homogeneously high level of digital competence across the sample.

The concern index was constructed by summing scores regarding positive or negative opinions on AI's impact on: (1) education, (2) future working careers, (3) society in

Table 1. Demographic and Academic Characteristics of the Sample (N = 1,366).

Characteristic	Category	n	%
Age	Mean	26	-
	SD	9.3	-
	Median	23	-
	Mode	19	-
Gender	Female	579	42.4
	Male	764	55.9
	Other	23	1.7
Academic Level	Undergraduate	862	63.1
	Master's	461	33.7
	Doctoral	43	3.1
Discipline	Industrial and Information Engineering	363	26.6
	Computing and ICT	275	20.1
	Natural Sciences	120	8.8
	Psychology	96	7
	Economics	93	6.8
	Literature-Humanities	83	6.1
	Education	55	4
	Other disciplines (< 4% each)	281	20.6

general, and (4) the risk that AI systems will eliminate many jobs in the long term. This produced a continuous index ranging from 4 to 16 ($M=9.76$; $Me=10.00$; $Mo=10.00$; $SD=2.88$). The mean and median located at the midpoint of the possible range, combined with a relatively substantial standard deviation, suggest a predominantly balanced distribution of concerns within the sample, indicating substantial inter-individual variability despite a moderate average level of concern.

K-means clustering procedure

To identify significant subgroups within the student population and understand different perspectives regarding artificial intelligence, a multivariate analysis based on the non-hierarchical K-means clustering technique was adopted. The clustering procedure was structured using variables deemed most relevant for differentiating students' views. In addition to the 'digital competence' and 'concern' indices described above, the following variables were included: (1) use of ChatGPT for personal purposes, (2) use of ChatGPT in support of study, (3) perception of AI's impact on future academic choices, (4) perception of AI's impact on postgraduate professional prospects, (5) opinions related to the statement 'students' use of ChatGPT defeats the purpose of education', (6) opinions related to 'the university should ban the use of ChatGPT', and (7) gender as a socio-demographic control variable.

The optimal number of clusters was determined using the Elbow method, based on a systematic evaluation of the within-cluster sum of squares (WCSS) across a range of K-means solutions. Specifically, K-means clustering was iteratively performed for solutions ranging from two to ten clusters, and for each solution the total within-cluster sum of squares was computed to quantify the residual variance. The inspection of the resulting WCSS values revealed a pronounced reduction up to the four-cluster solution (from 9,730.60 at $k=2$ –7,604.05 at $k=4$), followed by progressively smaller marginal improvements for higher values of k . Although the five-cluster solution further reduced the within-cluster sum of squares to 7,059.50, the associated gain in compactness (–544.5) was comparable to, yet slightly smaller than, the improvement observed when moving from three to four clusters (–633.9), while requiring an additional increase in model complexity. Beyond $k=5$, subsequent reductions in WCSS became increasingly modest and irregular, indicating the onset of diminishing returns. In light of the overall shape of the elbow curve and in accordance with the principle of parsimony, the four-cluster solution was therefore selected as the most appropriate compromise between explanatory adequacy and model simplicity.

The procedure thus identified four clusters, distributed as follows: Cluster 1 ($n=414$, 30.3%), Cluster 2 ($n=403$, 29.5%), Cluster 3 ($n=291$, 21.3%), and Cluster 4 ($n=258$, 18.9%). The chi-square test was employed to evaluate the statistical significance of observed differences, with cluster-by-variable cross-tabulation serving as the basis for this procedure (see [Table 2](#)).

Qualitative coding

The qualitative analysis of open-ended responses employed an inductive thematic coding approach (Thomas 2006). This approach was particularly suitable given the exploratory nature of the research and the relatively brief nature of most responses. The coding process unfolded in three phases. In phase 1, three researchers independently conducted a systematic reading of all responses concerning advantages and disadvantages of GAI, individually identifying and proposing preliminary codes based on recurring patterns

Table 2. Statistically significant variables sorted by cluster (%), and indexes. Response categories: Y = Yes; SA = Strongly Agree; EX = Excellent; AL = A lot; M = Male; F = Female; X = Other.

Key Sig. Vars by Cluster	Ans	FEAR	IND	PRAG	ENTH	χ^2	df	<i>p</i>
Used ChatGPT for Personal Projects	Y	54	54.7	79.7	78.7	95.68	3	.000
Used ChatGPT for University Tasks	Y	20.6	29.5	49.9	46.1	80.3	3	.000
Digital Competence	EX	11.3	3.5	22.8	27.3	203.15	9	.000
Smartphone Usage	AL	54	30.6	58.6	52.7	179.67	9	.000
Search Engines	AL	72.2	40.7	93.8	86	279.9	9	.000
Device Usage: Computer	AL	45.4	11.2	78.4	66.9	347.23	9	.000
Internet Service Usage: Social Network	AL	53.3	30.6	58.6	52.7	68.55	9	.000
ChatGPT generates better results than I can achieve on my own	SA	28.2	32.6	32.3	34.1	95.68	9	.000
University should regulate ChatGPT	SA	40.9	29.1	36.7	39.6	38.13	9	.000
Use of ChatGPT is morally wrong	SA	48.5	31.8	22.3	21.7	96.71	9	.000
ChatGPT generates results that can be mistaken for 'human'	SA	18.2	6.6	10.7	10.9	31.85	9	.000
ChatGPT has a non-transparent handling of user data	SA	30.2	15.5	16.6	12.6	79.73	9	.000
Use of ChatGPT defeats the purpose of the instruction	SA	36.1	19	7.2	7.5	246.16	9	.000
ChatGPT's importance is exaggerated by the media	SA	13.4	13.6	13.9	21.5	26.06	9	.002
University should ban ChatGPT	SA	16.2	9.7	1.2	2.4	96.71	9	.000
AI is influencing my choice of future study path	SA	16.4	9.7	26.2	24.9	78.26	9	.000
AI is influencing my choice of career path	SA	22	10.9	25.6	25.3	85.57	9	.000
AI will not be able to replace human intelligence or creativity	SA	88	74	82.5	88.4	30.98	9	.000
In the long run, AI systems will become the 'new normal'	SA	87.5	82.9	93.5	85.2	35.97	9	.000
In the long run, AI systems will take away many jobs	SA	95.3	69.7	82.4	66.8	627.57	9	.000
Concern for AI's Societal Impact	SA	79.7	19.8	21.8	2.9	246.16	9	.000
Gender	M	46.4	48.4	60.5	62.8	33.28	3	.000
	F	50.5	48.8	38.2	36.7	33.28	3	.000
	X	3.1	2.7	1.2	0.5	33.28	3	.000

and themes. In phase 2, the research team convened to compare proposed codes, discuss conceptual overlaps, and harmonize coding labels through consensus. This collaborative process resulted in a unified codebook comprising seven codes for advantages ('operational', 'informational', 'accessibility', 'creativity', 'education', 'professional', and 'other') and nine codes for disadvantages ('cognitive and learning risks', 'problematic behaviour', 'technical and reliability limitations', 'socio-cultural impacts', 'ethical issues', 'training issues', 'professional concerns', 'systemic/dystopian risks', and 'other'). In phase 3, each researcher independently applied the harmonized codebook to the entire dataset using NVivo software. The three coding sets were then systematically compared to identify discrepancies. While divergences were infrequent, researchers negotiated a consensus solution through discussion in all cases where they occurred, ensuring that the final coding reflected a shared interpretation.

Results

The cluster analysis revealed four distinct student profiles, each characterized by different levels of digital skills and attitudes toward GAI. Based on the principal characteristics that emerged within each group, we assigned synthetic labels: ‘the Indifferent’ (18.9%), ‘the Fearful’ (21.3%), ‘the Pragmatics’ (29.5%), and ‘the Enthusiasts’ (30.3%). These labels were chosen to be consistent with and representative of the main characteristics of each cluster, ordered by: (a) level of digital skills and (b) perception (positive or negative) regarding the advent of GAI (see [Figure 1](#)).

In the following subsections, we present findings organized by cluster, addressing each research question systematically. For each cluster, we first describe demographic and digital skill characteristics (addressing RQ1), then examine perceptions of GAI’s role in academic and personal activities (RQ2), perspectives on long-term implications (RQ3), and finally ethical and regulatory concerns (RQ4).

RQ1: patterns of adoption across socio-demographic factors and digital skills

The indifferent

The initial group, which we labeled ‘Indifferent’, constitutes 18.9% of the sample (n = 258). These students declared the lowest level of digital skills, with only 3.5% self-reporting ‘excellent’ digital skills – in stark contrast to the other three clusters. The demographic profile of this group is equally distributed between males and females. The Indifferent are less likely to use smartphones, computers, and tablets and demonstrate lower rates of use for online digital services, suggesting a certain detachment from the digital realm compared to their counterparts in other groups.

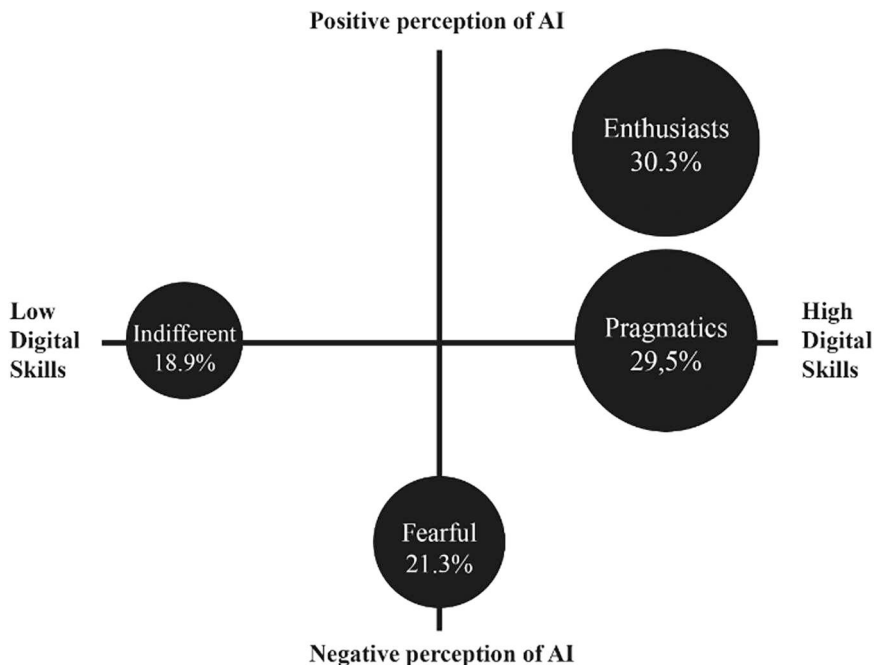


Figure 1. The four clusters by digital skills and AI perception.

Nevertheless, they are not inherently opposed to AI use. 29.5% indicated that they utilize ChatGPT to assist with university-related tasks, rising to 54.7% for personal projects. These figures are significantly lower than those recorded by the two ‘more digital’ clusters but at a similar rate to the Fearful cluster.

The fearful

The Fearful (21.3%, $n = 291$) possess intermediate digital skills between the Indifferent and the two more digital clusters. Their utilization of smartphones, computers, and digital services is situated between these extremes. Their social media engagement, however, aligns with the most digitally engaged clusters. Notably, this is the only group where females outnumber males.

They report the lowest GAI use for personal projects and significantly less GAI use for university-related tasks (20.6%) compared to the Indifferent (29.5%), the Pragmatics (49.9%), and the Enthusiasts (46.1%). Future AI adoption intentions are also markedly lower.

The pragmatics and the enthusiasts

These two clusters comprise 29.5% ($n = 403$) and 30.3% ($n = 414$) of the sample, respectively. In addition to being the most numerous clusters, they are predominantly male (the Pragmatics: 60.5%; the Enthusiasts: 62.8%). Their ‘excellent’ self-declared digital skills level is twice that of the Fearful and at least six times higher than the Indifferent. These clusters show the highest digital device and service usage (with the Pragmatics slightly exceeding the Enthusiasts) and exhibit remarkably similar GAI usage patterns for both personal and university-related activities, both currently and for anticipated future use (Figure 2).

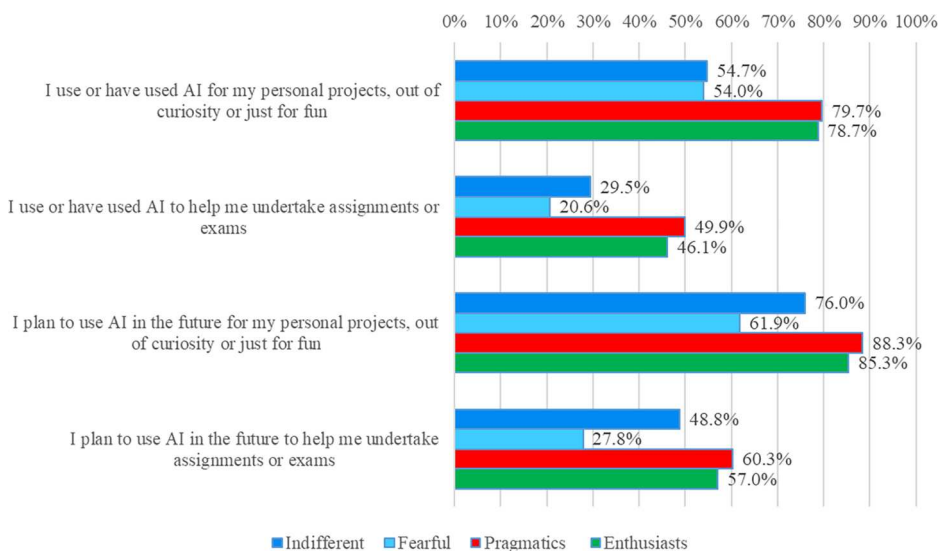


Figure 2. The four clusters and AI use for personal and university projects (present and future).

RQ2: perceptions of GAI's role in academic and personal activities*The indifferent*

This cluster expresses the weakest support for AI regulation. They occupy an intermediate position regarding: (1) whether using ChatGPT to perform tasks and exams is ‘morally wrong’, (2) whether AI use ‘defeats the purpose of education’, and (3) whether universities should prohibit its use. Furthermore, they are the least convinced that AI-generated results can be ‘mistaken for human’.

In open-ended responses regarding GAI's advantages and disadvantages (Tables 3 and 4), the Indifferent show low attention to ethical issues (3.1%, – 4 percentage points from the mean) but emphasize educational/training aspects (14%, +3.1). Their responses on disadvantages are often generic, with high selection of the ‘other (non-categorizable)’ option (9%, +3.1), suggesting a tendency toward vague answers, potentially due to lack of familiarity. Additionally, they underreport informational (11.4%, – 2.8) and operational (39.4%, – 2.1) advantages. This pattern indicates a detached view of AI, favoring educational aspects closer to their direct experience over ethical and technical considerations.

Table 3. Code list showing responses to advantages/disadvantages of AI (and typical responses).

Disadvantages	Typical Answer
Cogn. and learn. risks	AI may hinder independent thinking, reducing creativity/effort/skill development
Problematic behavior	Over-reliance on AI weakens engagement, reflection and personal idea development
Technical limits/reliability	AI may produce errors/inaccuracies; uncritical use weakens analysis skills
Socio-cultural impacts	Standardises thought, reduces diversity and weakens real-world social bonds
Ethical issues	Raises concerns re: plagiarism, cheating, data misuse and lack of transparency
Educational issues	Demotivates study, erodes core skills (e.g. writing, reasoning, autonomy)
Other (not categoriz.)	No clear experience/opinion on AI; some avoid or don't use it
Professional concerns	Threatens jobs, lowers value of human work, esp. in creative sectors
Systemic/dystopian risks	Undermining democracy and privacy by enabling disinformation, power concentration, surveillance and social manipulation
Advantages	Typical Answer
Operational advantages	AI speeds up/simplifies long/repetitive tasks, reducing time/effort (in study/in general).
Informative advantages	Quick, clear access to detailed info w/out checking multiple sources
Educational advantages	Helps grasp complex concepts and supports study via clear/personalized explanations
Accessibility advantages	Makes knowledge immediate and usable anywhere, even on complex topics
Creative advantages	Stimulates ideas and alternative views; aids project start and original development
Other (not categoriz.)	No use or no clear perceived benefits from AI
Professional advantages	Automates tasks and boosts productivity; frees time for strategic/creative work

Table 4. Coding of open-ended responses on perceived pros/cons of GAI tools. Cluster % and Δ from overall means. Bold indicates differences $\geq \pm 2$ pts.

Disadvantages	Media (%)	Fearful		Indifferent		Pragmatics		Enthusiasts	
		(%)	δ	(%)	δ	(%)	δ	(%)	δ
Cogn. and Learn. Risks	29.4	36.6	+7.2	31.3	+1.8	31.0	+1.5	20.8	-8.7
Problematic behavior	23.2	16.5	-6.7	21.4	-1.9	24.1	+0.9	29.3	+6.1
Tech limitations/reliability	15.4	7.9	-7.5	17.0	+1.6	14.1	-1.4	21.9	+6.5
Socio-cultural impacts	8.1	11.7	+3.6	9.9	+1.8	6.8	-1.2	4.9	-3.2
Ethical issues	7.1	7.2	+0.1	3.1	-4.0	9.4	+2.3	7.6	+0.5
Educational issues	6.4	6.7	+0.3	5.6	-0.8	4.9	-1.5	8.0	+1.6
Other (not categorizable)	5.8	5.7	-0.1	9.0	+3.1	2.7	-3.1	6.6	+0.8
Professional concerns	2.7	4.1	+1.4	2.2	-0.5	4.1	+1.5	0.6	-2.1
Systemic/dystopian risks	1.9	3.6	+1.7	0.6	-1.3	2.9	+1.0	0.4	-1.5
<i>Advantages</i>	(%)	(%)	δ	(%)	δ	(%)	δ	(%)	δ
Operational advantages	41.5	42.1	+0.6	39.4	-2.1	41.7	+0.3	42.6	+1.2
Informative advantages	14.2	15.0	+0.8	11.4	-2.8	17.1	+2.9	13.4	-0.9
Educational advantages	10.9	7.2	-3.7	14.0	+3.1	8.5	-2.3	13.8	+2.9
Accessibility advantages	10.7	11.0	+0.2	10.5	-0.2	12.4	+1.7	9.0	-1.7
Creative advantages	9.4	6.6	-2.8	10.2	+0.7	11.8	+2.4	9.0	-0.4
Other (not categorizable)	10.2	15.0	+4.8	12.1	+1.9	5.6	-4.6	8.1	-2.1
Professional advantages	3.1	3.2	+0.1	2.5	-0.6	2.7	-0.4	4.1	+1.0

The fearful

This group most strongly supports GAI regulation in universities and is most convinced that GAI use in university is ‘morally wrong.’ Unsurprisingly, they are also most convinced that universities should prohibit AI use and that ChatGPT deployment by students ‘defeats the purpose of education’.

Their AI concerns extend to transparency, as they are the only group explicitly sceptical about ChatGPT’s data collection practices. This seriousness – contrasting with the Indifferent – is evident in their belief that AI outputs can be ‘mistaken for human’. They share with the Indifferent the lowest endorsement of AI’s potential positive consequences on human abilities, but express potential negative consequences to a much greater extent than all other groups. They are also most convinced that AI cannot replace human intelligence or creativity.

The Fearful cluster’s risk perception is strongly oriented toward cognitive risks (36.6%, +7.2) and socio-cultural impacts (11.7%, +3.6). In contrast, problematic behavior (16.5%, - 6.7) and technical limitations (7.9%, - 7.5) are under-represented, potentially due to lesser GAI use and thus less specific knowledge. Regarding benefits, this group struggles to identify specific advantages, as seen in the high proportion of the ‘other’ category (15%, +4.8) and low recognition of educational (7.2%, - 3.7) and creative (6.6%, - 2.8) benefits.

The pragmatics and the enthusiasts

These clusters show remarkably similar patterns regarding the belief that AI products can be ‘mistaken for human’. Moral, regulatory, and prohibition-related dimensions follow the

same pattern: ‘the university should identify ways to regulate the use of GAI’ (strongly agree: 36.7%; 39.6%); ‘the use of ChatGPT for tasks and exams is morally wrong’ (22.3%; 21.7%); ‘ChatGPT use defeats the purpose of education’ (7.2%; 7.5%); ‘the university should ban ChatGPT’ (1.2%; 2.4%).

The primary distinction lies in the extent of outright rejection of negative statements, consistently higher among the Enthusiasts. For example, ‘universities should ban GAI’ is strongly disagreed with by 66.7% of the Enthusiasts versus 58.6% of the Pragmatics. Similarly, ‘AI defeats the purpose of education’ is strongly rejected by 37.7% of the Enthusiasts compared to 27.5% of the Pragmatics.

In open-ended responses, the Pragmatics exhibit a distinct profile, emphasizing informational (17.1%, +2.9) and creative (11.8%, +2.4) benefits while providing fewer generic responses. They also show notable focus on ethical considerations (9.4%, +2.3), suggesting a more nuanced perspective that identifies specific advantages and disadvantages rather than generic evaluations.

The Enthusiasts present an intriguing risk perception profile, with significant underrepresentation of cognitive risks (20.8%, – 8.7) but notable emphasis on risks related to problematic behavior (29.3%, +6.1) and technical limitations (21.9%, +6.5), possibly reflecting greater awareness from frequent AI use. There is marked under-representation of socio-cultural impacts (4.9%, – 3.2), while more educational/training benefits are stated (13.8%, +2.9), indicating a more practical and technical view of AI.

RQ3: perspectives on long-term implications for career, academic growth, and society

The indifferent

This cluster is the least attached to AI. While they use it more than the Fearful, they assign it little significance. Among the least digital respondents, they feel somewhat alienated from these developments, perceiving AI as abstract and distant. They hold no strong expectations or fears regarding GAI and do not recognize its impact on their lives.

The fearful

Although they report minimal GAI impact on their educational and professional choices (similar to more digital groups), their serious stance on GAI’s effects on education, careers, society, and employment is evident across all dimensions (see [Figure 3](#) for cluster comparison of concerns). They share with the Indifferent the lowest agreement regarding AI’s potential positive consequences on human abilities, but express potential negative consequences on human abilities to a much greater extent than all other groups.

The pragmatics and the enthusiasts

The impact of GAI on human abilities (regarding potential beneficial outcomes) follows a similar pattern for both clusters, with slightly higher positive perceptions among the Enthusiasts. However, regarding negative effects, the most pronounced divergences emerge. The Pragmatics demonstrate percentages of concern regarding negative effects of GAI on human abilities analogous to those of the Indifferent, considerably higher than the Enthusiasts, particularly regarding ‘writing’, ‘reading’, ‘thinking’, and ‘stimulating creativity’ ([Table 5](#)).

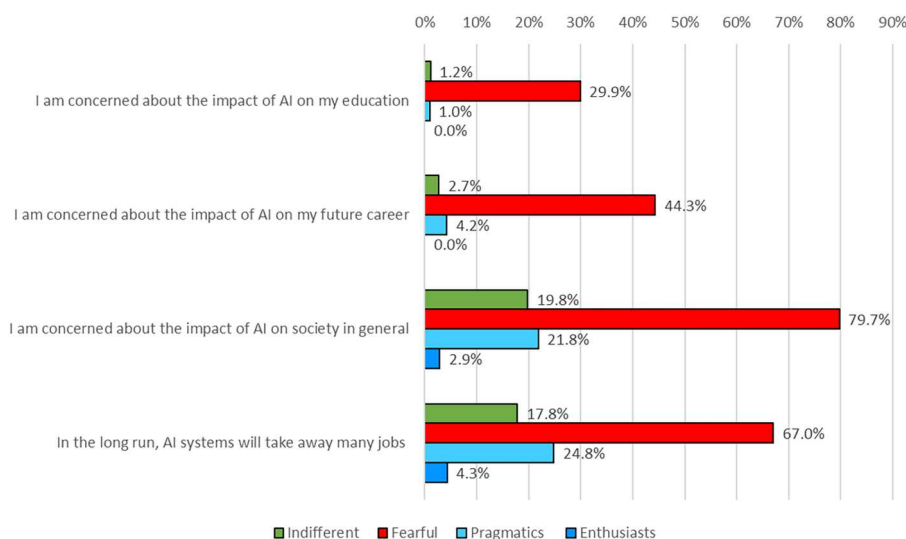


Figure 3. The four clusters and concerns towards AI ('strongly agree' responses).

The distinction between these groups becomes evident when examining views on AI's influence on human life and related concerns. The belief that 'AI is influencing' study and career choices is held by a minority in both clusters with similar proportions (strongly agree: 8.9% vs. 7.4%). However, differences are notable in outright rejection of this idea (strongly disagree: 39.5% vs. 58.7%). The Pragmatics are less likely to dismiss AI's influence on study choices (39.5%) compared to the Enthusiasts (58.7%). Likewise, rejection of AI's influence on career decisions is lower among the Pragmatics (36.2%) than the Enthusiasts (58%).

An even sharper divergence emerges regarding concerns. Only 1% of the Pragmatics and 0% of the Enthusiasts strongly agree with 'I am concerned about AI's impact on my education', whereas rates of strong disagreement differ significantly (strongly disagree: 25.6% vs. 80.2%). Similarly, none of the Enthusiasts strongly agree that 'AI affects my future career' (compared to 4.2% of the Pragmatics), yet total rejection is far higher among the Enthusiasts (76.8%) than the Pragmatics (10.7%). Moreover, apprehension regarding prospective adverse repercussions of AI on society is endorsed (strongly agree) by 21.8% of the Pragmatics versus only 2.9% of the Enthusiasts.

RQ4: ethical and regulatory concerns regarding GAI use

The indifferent

The Indifferent express the weakest support for AI regulation and occupy an intermediate position on whether using ChatGPT for academic tasks is morally wrong or defeats educational purposes. In open-ended responses, they show low attention to ethical issues (3.1%, – 4 percentage points), with responses often being generic.

The fearful

This group most strongly supports GAI regulation in universities and most strongly believes that GAI use is 'morally wrong.' They are most convinced that universities

Table 5. Clusters and perceived AI impact on future human abilities (isolated ‘Very Positive’ and ‘Very Negative’ responses).

Very Positive	Search Info	Writing	Reading	Thinking	Stim. Crea.	Understanding	Crit. Think.
Indifferent	19.4%	4.7%	2.7%	1.9%	7.0%	13.2%	3.9%
Fearful	18.2%	4.8%	5.8%	2.1%	4.5%	11.0%	2.7%
Pragmatics	30.0%	12.2%	7.4%	5.5%	11.9%	21.1%	4.0%
Enthusiasts	34.1%	16.9%	9.9%	6.3%	11.8%	25.8%	8.0%
Very Negative	Search Info	Writing	Reading	Thinking	Stim. Crea.	Understanding	Crit. Think.
Indifferent	3.9%	11.2%	8.9%	25.6%	24.0%	7.8%	21.3%
Fearful	9.3%	27.1%	22.3%	48.1%	41.6%	23.4%	38.5%
Pragmatics	3.5%	10.9%	7.4%	21.6%	20.6%	5.2%	15.9%
Enthusiasts	2.9%	4.3%	3.9%	14.3%	11.8%	4.1%	13.8%

should prohibit AI use. Their concerns extend to transparency and data collection practices, being the only group explicitly sceptical about ChatGPT's handling of user data. The Fearful cluster emphasizes ethical issues in their open responses, though their lower GAI usage may limit specific technical knowledge.

The pragmatics and the enthusiasts

Both clusters show moderate support for regulation (strongly agree: 36.7% and 39.6% respectively) but differ markedly in their rejection of restrictive measures. The Enthusiasts more strongly reject the notion that universities should ban ChatGPT (66.7% strongly disagree vs. 58.6% for the Pragmatics) or that AI defeats educational purposes (37.7% vs. 27.5%).

In open-ended responses, the Pragmatics show notable focus on ethical considerations (9.4%, +2.3), suggesting a balanced perspective that acknowledges both benefits and ethical challenges. The Enthusiasts, while showing less concern about cognitive and socio-cultural impacts, demonstrate greater awareness of technical limitations and problematic behaviors, likely reflecting their more extensive usage experience.

The distinct labels assigned to these clusters highlight their differences. For the Pragmatics, GAI is a resource integral to their lives, but they do not ascribe any particular 'messianic' expectations to it and approach GAI with a degree of objectivity and balance, intermittently highlighting positive or negative aspects. In contrast, the Enthusiasts perceive GAI as entirely positive, with no negative effects. They believe it will enhance all human abilities and do not associate it with any concerns or fears, whether personal or social. The Fearful view GAI with significant apprehension, emphasizing risks over benefits and advocating for institutional controls. The Indifferent remain largely disengaged, neither embracing GAI enthusiastically nor expressing strong concerns about its implications.

Table 6 provides a comprehensive summary of the main characteristics across the four clusters, synthesizing findings related to all four research questions.

Conclusion

Through a large-scale survey of 1,366 students from 24 Italian universities, this study reveals that engagement with Generative AI is neither uniformly embraced nor rejected. Rather, students navigate a complex terrain where digital competency, disciplinary context, gender, and ethical reasoning create four distinct profiles: the Indifferent (18.9%), the Fearful (21.3%), the Pragmatics (29.5%), and the Enthusiasts (30.3%). These profiles differ substantially in digital skills and attitudes toward AI adoption. Critically, the divide is not simply between users and non-users, but between those who approach GAI with nuanced understanding of both its affordances and limitations (the Pragmatics) and those at the extremes, either enthusiastically uncritical (the Enthusiasts) or fearfully resistant (the Fearful). The Indifferent, meanwhile, remain largely disengaged from these debates despite operating in increasingly AI-mediated academic environments. The findings align with research conducted internationally, which indicates a heterogeneous student response, encompassing both interest and apprehension, towards GAI in education (e.g. Malmström, Stöhr, and Ou 2023; Meade, Ogundamisi, and Tiratelli 2023). Cross-national comparisons indicate a degree of divergence in student perceptions of GAI; however, overarching themes, including the necessity of regulatory frameworks,

Table 6. Summary of the main characteristics of the 4 clusters.

Dimension	Indifferent	Fearful	Pragmatics	Enthusiasts
Digital skills	Lowest among all clusters; minimal technological engagement	Intermediate; moderate digital skills but lower AI literacy	High; strong digital proficiency and frequent technology use	Highest; highly proficient and comfortable with AI technologies
Attitude toward AI	Neutral or disengaged, with limited enthusiasm or concern	Sceptical and apprehensive, perceiving AI as potentially harmful	Balanced, acknowledging both benefits and challenges	Strongly optimistic, rejecting concerns about AI's risks
Perceived moral concerns about AI use in academia	Moderate concerns, but not strongly opposed	Highest moral concerns; view AI-assisted academic work as unethical	Few moral concerns; acknowledge ethical considerations but favor responsible use	Minimal to no moral concerns; advocate for full AI integration in academia
Support for university regulation of AI	Weakest support for AI regulations	Strongest support for institutional control and ethical oversight	Moderate support; advocate for responsible but flexible AI policies	Least support; prefer integration over restriction
Perceived impact of AI on human abilities	Unconcerned; do not foresee significant consequences	Predominantly negative; fear cognitive decline and loss of critical thinking skills	Mixed perspective; recognize benefits but are cautious of potential drawbacks	Unquestionably positive; believe AI will enhance all human abilities
Perceived long-term risks of AI	Limited awareness or concern about AI's societal consequences	Strong concerns about AI's impact on education, employment, and social structures	Moderate concerns; acknowledge risks but remain pragmatic	Minimal concerns; reject the idea that AI will negatively affect society
Influence of AI on academic and career decisions	No significant influence on study or career choices	Negative; fear job displacement and decline in educational integrity	Open to AI's role in shaping career prospects and educational pathways	Dismissive of AI's impact on personal career and academic choices
Concerns about AI-related job displacement	Moderate; acknowledge potential job market shifts but not alarmed	Strongest concerns; view AI as a major threat to employment	Moderate; aware of potential changes but focus on adaptability	Unconcerned; believe AI will create more opportunities than it eliminates
View on AI as 'the new normal'	Sceptical about AI's long-term presence in daily life	Strong belief that AI will disrupt traditional societal structures	Accept AI as an integral part of the future but with caution	Confident that AI will seamlessly integrate into all aspects of life
Concerns about AI's ethical risks (e.g. misinformation, bias, privacy issues)	Limited awareness of ethical implications	Highest concerns; sceptical of AI's reliability and transparency	Moderate concerns; recognize ethical risks but focus on mitigation strategies	Least concerned; perceive ethical risks as negligible compared to AI's benefits

ethical considerations, and the cultivation of AI literacy, appear to be central determinants influencing student interaction with these technologies.

Our findings challenge simplistic narratives. High digital competence does not automatically translate into critical AI literacy. The Enthusiasts, despite their technical proficiency, demonstrate the lowest awareness of cognitive risks and ethical concerns. This suggests that this group may be most susceptible to the ‘cognitive debt’ identified by Kosmyna et al. (2025), as their high usage is coupled with low risk awareness. Conversely, the Fearful’s apprehensions, while sometimes limiting their engagement, reflect legitimate concerns about academic integrity, cognitive development, and the erosion of human-centred learning. The gender disparity observed across clusters (with males dominating the Pragmatics and the Enthusiasts at approximately 60%) and the disciplinary variations (with STEM students showing higher engagement) underscore that GAI adoption is embedded within existing inequalities and epistemological traditions. These patterns suggest that without intentional intervention, GAI integration risks reproducing and potentially amplifying existing educational disparities, particularly affecting female students and those in humanities disciplines.

In light of the findings, it is clear that the relationship between students and GAI is complex and evolving, influenced by individual experiences, digital skills, and specific academic curricula. The observed gender disparity suggests the need for targeted interventions to ensure equitable GAI adoption across underrepresented groups. Furthermore, variations in GAI use across different disciplines underscore the importance of customized AI training and support tailored to each field. Efforts should be made to ensure equal opportunities for all students to engage with GAI. Moreover, the significant concerns expressed by some students emphasize the necessity for establishing clear ethical guidelines to alleviate anxieties surrounding AI’s role in academia.

The tension between innovation and integrity emerges as the central challenge facing higher education institutions. The research suggests that outright bans on GAI use are unlikely to be effective or beneficial. Instead, universities should develop and communicate clear policies regarding AI use in coursework, exams, and research, addressing issues of attribution, plagiarism, and academic integrity. They must also establish robust mechanisms to identify and prevent AI-related misconduct. However, the uncritical enthusiasm demonstrated by some students equally threatens the core educational values of authentic intellectual engagement and original thought. Universities must therefore chart a middle course that acknowledges GAI as an integral component of contemporary knowledge work while establishing clear ethical frameworks and fostering critical AI literacy across all student populations.

To achieve this balance, educational institutions should incorporate GAI literacy courses or workshops into curricula to educate students on ethical GAI use and critical evaluation of AI-generated content. Discipline-specific GAI training and resources should be provided to highlight relevant applications and potential career impacts. Educators should foster discussions on GAI ethics, guiding students in evaluating AI’s role in their learning and future careers. Assignments should encourage critical analysis of AI-generated content to develop strong analytical and reasoning skills. These interventions must specifically target underrepresented groups (particularly female students and humanities majors) to prevent the crystallization of AI engagement gaps.

Collaboration between students and universities should be encouraged. Educators should work closely with students to explore innovative and ethical ways of integrating GAI into learning, rather than imposing restrictive policies. Creating institutional spaces for ongoing dialogue among students, faculty, and administrators about evolving

GAI practices ensures that policies remain responsive to technological developments and student needs rather than becoming static prohibitions divorced from lived realities.

These recommendations, while grounded in the Italian context, have broader implications for higher education systems worldwide. The four profiles we identified likely exist across national contexts, though their proportions and specific characteristics may vary based on cultural norms, regulatory environments, and institutional infrastructures. Understanding these profiles enables more targeted and effective interventions than one-size-fits-all approaches.

Several critical questions emerge for future research. Longitudinal studies are essential to track how student profiles evolve as GAI becomes normalized in academic settings: Do the Indifferent become engaged? Do Enthusiasts develop more critical perspectives through experience? Do institutional interventions successfully shift students toward pragmatic engagement? Cross-national comparative research examining how different regulatory environments and cultural contexts shape student engagement patterns would illuminate which institutional approaches most effectively balance innovation with integrity. Experimental studies testing the effectiveness of specific interventions, such as AI literacy workshops, discipline-specific training modules, or collaborative AI-human assignments, could provide evidence-based guidance for pedagogical practice. Investigation of the ‘cognitive debt’ phenomenon identified by Kosmyna et al. (2025) deserves particular attention, especially among Enthusiasts who may be developing dependencies that undermine deep learning without recognizing these effects. Finally, research into the long-term career and educational outcomes associated with different GAI engagement patterns could help validate or challenge current assumptions about the necessity of AI skills for professional success.

The broader societal implications extend beyond educational contexts. As GAI technologies permeate professional environments, the critical thinking skills, ethical reasoning capabilities, and ability to distinguish between AI-assisted and AI-substituted work that we seek to cultivate in university students will become essential competencies for informed citizenship and responsible innovation. Universities, as sites where young adults develop not only technical skills but also intellectual habits and ethical commitments, bear particular responsibility for shaping how the next generation of professionals, policymakers, and citizens engage with AI technologies. By implementing evidence-based policies that recognize the diversity of student perspectives and needs, universities can foster responsible GAI integration that enhances rather than undermines educational quality. The goal is not to produce either uncritical enthusiasts or fearful resisters, but rather thoughtful pragmatists who can leverage AI tools critically and ethically while maintaining authentic intellectual engagement. This requires moving beyond binary choices of permission or prohibition toward more nuanced approaches that integrate ethical reasoning, technical understanding, and disciplinary expertise into a comprehensive framework for AI literacy in higher education. Only through such multifaceted efforts can we prepare students for a future where AI is neither threat nor panacea, but rather a powerful tool requiring informed, critical, and ethically grounded human judgment.

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Author contributions

CRediT: **Claudio Melchior**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Manuela Farinosi**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing.

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Ethics statement

All procedures were conducted in accordance with the ethical standards set forth by the institutional and national research committee and with the 1964 Helsinki Declaration and its subsequent amendments or comparable ethical standards. Approval was granted by the Institutional Review Board of the Department of Mathematics, Computer Science and Physics of the University of Udine (approval date: 26/09/2023).

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